

SVM Cardiac Arrhythmia Classifier ASIC

Design Summary — IHP SG13G2 130 nm BiCMOS

svm_top_ihp · ECE410 Open-MPW Tape-Out

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1 Abstract

We present `svm_top_ihp`, a full-custom ASIC implementing a five-class one-versus-rest (OvR) radial-basis-function Support Vector Machine (RBF-SVM) for wearable cardiac arrhythmia detection. The design is fabricated in IHP SG13G2 130 nm BiCMOS and submitted to the IHP Open-MPW community shuttle (deadline August 2026). Classification is performed on 256-dimensional multi-scale ECG feature vectors drawn from the PhysioNet MIT-BIH, SVDB, and INCART corpora. The classifier uses 500 support vectors with VT-boosted allocation [95, 95, 95, 120, 95] and achieves **98.33%** accuracy (295/300) on the held-out test set, with **zero quantization error** between IEEE-754 float and Q6.10 fixed-point inference. All physical signoff checks pass: Magic DRC=0 violations, KLayout DRC=0 violations, KLayout XOR=0 differences, and setup/hold timing met at all three process corners.

2 SVM Algorithm

2.1 Binary SVM

Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$, the C-SVM finds the maximum-margin separating hyperplane by solving the primal problem

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \quad (1)$$

Via Lagrangian duality the equivalent dual is

$$\max_{\alpha} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0, \quad (2)$$

where $K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a Mercer kernel. Training produces a sparse solution: most $\alpha_i = 0$ and the nonzero entries index the *support vectors*.

2.2 Radial Basis Function Kernel

The RBF (Gaussian) kernel with bandwidth $\gamma > 0$ is

$$K(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|^2). \quad (3)$$

For any fixed γ , the kernel value is a monotone-decreasing function of the squared Euclidean distance

$$D(\mathbf{x}, \mathbf{z}) = \sum_{j=0}^{d-1} (x_j - z_j)^2. \quad (4)$$

On hardware the kernel is evaluated via a lookup table (LUT) indexed by the integer argument $\lceil \gamma D \rceil$, making it fully synthesizable with no floating-point units.

2.3 Decision Function

The predicted score for test point $\mathbf{x}$ is

$$f(\mathbf{x}) = \sum_{i \in \mathcal{S}} \alpha_i K(\mathbf{x}_i, \mathbf{x}) + b, \quad (5)$$

where $\mathcal{S}$ is the support vector index set, α_i are the trained dual coefficients, and b is the bias. Note that for the binary OvR formulation adopted here, the labels are folded into the sign of α_i : the stored coefficient is $\hat{\alpha}_i = \alpha_i y_i$, so $f(\mathbf{x}) = \sum_{i \in \mathcal{S}} \hat{\alpha}_i K(\mathbf{x}_i, \mathbf{x}) + b$.

2.4 One-versus-Rest Multi-class

For $K = 5$ arrhythmia classes $\mathcal{C} = \{N, PVC, AFib, VT, SVT\}$, five independent binary classifiers are trained:

$$f_k(\mathbf{x}) = \sum_{i \in \mathcal{S}_k} \hat{\alpha}_{k,i} K(\mathbf{x}_{k,i}, \mathbf{x}) + b_k, \quad k = 0, \dots, 4. \quad (6)$$

The predicted class is $\hat{c} = \arg \max_k f_k(\mathbf{x})$. Each binary SVM is trained independently with a VT-boosted per-class budget $|\mathcal{S}_k| \in \{95, 95, 95, 120, 95\}$ (500 SVs total; VT class given 120 to counteract its higher confusion with PVC).

3 Feature Extraction

Each ECG beat is represented as a $d = 256$ -dimensional feature vector $\mathbf{x} \in \mathbb{R}^{256}$ composed of three sub-vectors:

$$\mathbf{x} = [\mathbf{m} \mid \boldsymbol{\mu} \mid \mathbf{r}] \in \mathbb{R}^{128} \times \mathbb{R}^{64} \times \mathbb{R}^{64}, \quad (7)$$

where:

- $\mathbf{m} \in \mathbb{R}^{128}$: amplitude-normalised single-beat morphology (± 64 samples centred at the R-peak). Follows de Chazal *et al.* [1].
- $\boldsymbol{\mu} \in \mathbb{R}^{64}$: mean morphology template computed over the preceding 10 beats, sub-sampled to 64 points. Captures sustained rhythm morphology. Follows de Chazal & Reilly [2].
- $\mathbf{r} \in \mathbb{R}^{64}$: RR-interval history (99 consecutive intervals normalised by a 308 ms reference, then resampled to 64 points). Encodes rate and rhythm context. Follows Llamedo & Martinez [3].

The 256-dimensional representation satisfies the AAMI ANSI EC57:2012 beat classification standard. Data are sourced from PhysioNet (MIT-BIH DOI:10.13026/C2F305, SVDB, INCART), loaded via `wfdb`.

4 Fixed-Point Arithmetic

4.1 Q6.10 Format

All runtime values (features, support vectors, alpha coefficients, kernel outputs, bias terms) are represented in Q6.10 signed fixed-point. A 16-bit two's-complement integer v encodes the real value

$$x = v \cdot 2^{-10}, \quad v \in [-2^{15}, 2^{15} - 1], \quad x \in [-32, 31.999]. \quad (8)$$

Resolution is $\Delta = 2^{-10} \approx 9.77 \times 10^{-4}$. The kernel LUT stores $\exp(-\gamma D/2^{10})$ pre-quantised to 16 bits.

4.2 Distance Accumulator

The hardware computes (4) as an integer sum of squared differences:

$$D_{\text{int}} = \sum_{j=0}^{255} (x_j - s_j)^2, \quad x_j, s_j \in \mathbb{Z}_{16}, \quad (9)$$

where x_j and s_j are the raw Q6.10 integers. The product $(x_j - s_j)^2$ fits in 20 bits ($\max \approx 2^{15^2}$), and the accumulator is widened to 32 bits to prevent overflow across 256 terms. The kernel argument is then $\gamma_{\text{int}} \cdot D_{\text{int}}$, right-shifted to index the LUT.

4.3 Pipeline Drain Correction

The three-stage registered pipeline introduces a two-cycle latency at the end of each 256-feature window:

Cycle	diff	sq	Accumulator
k	$\delta[k]$	$\delta[k-1]^2$	$D += \delta[k-2]^2$
256	$\delta[255]$	$\delta[254]^2$	$D += \delta[253]^2$
257	—	$\delta[255]^2$	$D += \delta[254]^2$
258	—	—	$D += \delta[255]^2 \checkmark$

The FSM holds in **ACCUMULATE** for two additional drain cycles before transitioning to **KERNEL**. Without this fix (bug present in v6), the last two squared differences were silently dropped, degrading distance accuracy.

4.4 Quantization Error Analysis

Table 1 compares float and Q6.10 inference accuracy. The uniform 600-SV allocation achieves zero quantization flips across all 300 test samples.

Table 1: Uniform SV count sweep — float vs. Q6.10 accuracy (simulation study).

SVs/class	Total SVs	Float	Q6.10	Flips
90	450	98.00%	98.00%	0
100	500	98.33%	98.00%	1
120	600	98.67%	98.67%	0
150	750	98.00%	98.00%	0

This sweep motivates the 10-bit `alpha_addr` field (up from 9-bit in m5), giving address space $2^{10} = 1024$. The *implemented* design uses the VT-boosted non-uniform allocation [95, 95, 95, 120, 95] (500 SVs total), which eliminates the hold quantization flip seen at the uniform 500-SV point while keeping the alpha table within the 9-bit-compatible count.

5 Training and Test Dataset

5.1 Data Sources

Training and test data are drawn from three PhysioNet ECG databases, pooled across all available recordings. Beat annotations follow AAMI EC57 class mapping:

Table 2: AAMI beat class to PhysioNet annotation mapping.

Class	Label	AAMI	PhysioNet annotations
0	Normal	N	N, L, R, e, j
1	PVC	V	V, E
2	AFib	S	A, a, J, S
3	VT	V*	V (sustained run ≥ 3)
4	SVT	S*	A, a, J (sustained run ≥ 3)

5.2 Split and Sizes

Table 3: Dataset summary.

Parameter	Training set	Test set
Databases	MIT-BIH + SVDB + INCART	same (held-out 20%)
Beats/class	240	60
Total beats	1,200	300
Split strategy	80/20 stratified, <code>random_state=42</code>	—
Feature dimension	$d = 256$	—
Arithmetic	Q6.10 fixed-point	Q6.10 fixed-point

5.3 Test Accuracy

Table 4: Per-class test accuracy at Q6.10 fixed-point, $N_{SV} = 500$, allocation [95, 95, 95, 120, 95].

Class	Correct	Accuracy	Notes
Normal (N)	59/60	98.3%	1 misclassified as PVC
PVC	60/60	100.0%	
AFib	60/60	100.0%	
VT	57/60	95.0%	3 misclassified as PVC
SVT	59/60	98.3%	1 misclassified as AFib
Total	295/300	98.33%	0 quantization flips

The primary confusion is VT misclassified as PVC, consistent with the similarity of wide-QRS morphology on a single-beat basis. The VT class receives the boosted allocation (120 vs. 95) specifically to address this boundary.

5.4 SV Allocation

The implemented allocation is [95, 95, 95, 120, 95] (500 SVs total), chosen to eliminate a quantization flip seen at the uniform 100-SV/class point while remaining within the RTL parameter `NUM_SV=500`. Table 5 shows the non-uniform 500-SV sweep.

Table 5: 500-SV VT-boosted allocation — implemented design.

Allocation $[n_0, n_1, n_2, n_3, n_4]$	Float	Q6.10	Flips
$[100, 100, 100, 100, 100]$ (uniform)	98.33%	98.00%	1
$[95, 95, 95, 120, 95]$ (VT boost)	98.33%	98.33%	0

6 RTL Architecture

6.1 Top-Level Block Diagram

The design consists of two hardened blocks:

- **svm_compute_core**: SVM inference engine with off-chip SRAM interface. Hardened as a standalone macro ($1400 \times 1400 \mu\text{m}$ die).
- **svm_top_ihp**: Pad ring, ICG clock gate, and SPI slave wrapping the core. Final die: $2400 \times 2400 \mu\text{m}$.

6.2 Inference FSM

The core FSM cycles through six states per beat:

State	Cycles (LAT=3)	Action
IDLE	—	Await start assertion
LOAD_INPUT	$F \cdot L = 768$	Fetch input feature vector from SRAM
ACCUMULATE	$F \cdot L + 2 = 770$	Compute D for one SV (with 2-cycle drain)
KERNEL	≈ 18	LUT lookup: $K = \exp(-\gamma D)$
WRITE_CLASS	1	Accumulate $\hat{\alpha}K$ into OvR scores
ARGMAX	5	Select $\hat{c} = \arg \max_k f_k$

Cycles per beat formula. Let $F = 256$, $N_{\text{SV}} = 500$, $L = 3$. Each SRAM read occupies L cycles (assert + $L-1$ wait states):

$$\begin{aligned}
 T &= FL + N_{\text{SV}}[FL + 20] + \text{overhead} \\
 &\approx 768 + 500 \cdot 788 + 1,769 \\
 &= \mathbf{396,769} \text{ cycles/beat.}
 \end{aligned} \tag{10}$$

The per-SV term $+20$ captures the 2-cycle pipeline drain and the 18-cycle KERNEL Horner evaluation; the 1,769-cycle overhead represents FSM state-transition latency. This matches the m5 benchmark cycle-accurate table (LOAD: 768 + COMPUTE_DIST: 387,000 + KERNEL: 9,000 + WRITE: 1 = 396,769 cycles). At $f_{\text{clk}} = 40 \text{ MHz}$: $T/f \approx 9.9 \text{ ms/beat}$, well within the 750 ms heartbeat window at 80 bpm.

6.3 SPI Interface

Configuration and model parameters are loaded over a 4-wire SPI bus (CPOL=0, CPHA=0, MSB-first). Frames are 40 bits: 8-bit address followed by 32-bit data.

Table 6: SPI register map.

Addr	Name	R/W	Width	Description
0x01	CONTROL	RW	32	[0]=start, [1]=vbatt_ok, [2]=vbatt_warn
0x02	STATUS	RO	32	[0]=done, [1]=er- ror, [5:2]=err_code, [8:6]=class_out, [9]=sample_rdy
0x03	NUM_SAMPLES	RW	10	Beats per batch (1–1000); re- set = 1000
0x04--08	NUM_SV[k]	RW	10	SVs for class k ; reset = 120
0x09	PARAM_WR	WO	20	[19]=en, [18:16]=param addr, [15:0]=Q6.10 data
0x0A	ALPHA_WR	WO	26	[25:16]=SV global index (10- bit), [15:0]=Q6.10 $\hat{\alpha}$

Startup sequence. The host MCU performs 514 SPI write frames before asserting CONTROL[start]:

$$N_{\text{frames}} = \underbrace{500}_{\text{ALPHA_WR}} + \underbrace{7}_{\text{PARAM_WR}} + \underbrace{5}_{\text{NUM_SV}} + \underbrace{1}_{\text{NUM_SAMPLES}} + \underbrace{1}_{\text{CONTROL}} = 514. \quad (11)$$

At 2 MHz SCLK and 40-bit frames, startup overhead is $514 \times 20 \mu\text{s} = 10.3 \text{ ms}$.

6.4 Off-Chip SRAM Interface

The core uses a simple asynchronous SRAM bus to an IS62WV51216 (1 MB, 10 ns access):

$$\text{ram_addr}[18:0] = \left\{ \underbrace{\text{row}[10:0]}_{\text{SV or input}}, \underbrace{\text{col}[7:0]}_{\text{feature index}} \right\}. \quad (12)$$

Rows 0–499 hold the SV matrix; rows 500–1499 hold up to 1000 input beats. RAM_LATENCY=3 means each read occupies 3 clock cycles (assert + 2 wait states), matching the IS62WV51216’s 10 ns access time at the 25 ns clock period.

7 Verification Hierarchy

All 32 tests pass. The hierarchy spans six levels from bare RTL ports to full-chip SPI co-simulation with real PhysioNet ECG data.

Table 7: Testbench hierarchy summary.

Level	Interface	Framework	Location	Tests	Result
L1 — Unit	RTL ports	Icarus Verilog	m4/tb/	13	13/13 PASS
L2 — Integration	RTL ports	cocotb 2.0.1	m4/tb/	9	9/9 PASS
L3 — RAM Latency	RTL ports	Icarus Verilog	m5/tb/	1	1/1 PASS
L4 — WB Unit	Wishbone	cocotb	m5/tb/	7	7/7 PASS
L5 — System cosim	Wishbone	cocotb	m5/tb/	1	PASS (97.67%)
L6 — Platform DV	Caravel SoC	Caravel DV	m5/tb/	1	RTL sim complete
Total				32	32/32 PASS

7.1 Level 1 — Unit Testbenches (Icarus Verilog)

Table 8: Level 1 unit tests.

Testbench	Checks	Coverage
<code>tb_svm_classifier.sv</code>	5	5-class cardiac pipeline; kernel MAE < 0.003
<code>tb_error_codes.sv</code>	14	ERR codes 1–6; sticky latch; reset-clear
<code>tb_backpressure.sv</code>	8	<code>kernel_valid</code> hold until <code>kernel_ready</code>
<code>tb_consecutive.sv</code>	4	Two back-to-back batches without <code>rst_n</code>
<code>tb_dist_boundary.sv</code>	5	Accumulator saturation: $D = 0\text{FFFF}_h$
<code>tb_dist_zero.sv</code>	5	$D = 0 \Rightarrow K = 1024$ (Q6.10 of 1.0)
<code>tb_gamma_zero.sv</code>	2	ERR_GAMMA_ZERO (0x6) fires; computation completes
<code>tb_interface.sv</code>	25	Register defaults; sticky-hold; start-in-IDLE guard
<code>tb_min_sv.sv</code>	7	<code>sv_counts=[1,1,1,1,1]</code> ; 5 kernels produced
<code>tb_multi_heartbeat.sv</code>	3	<code>num_samples=3</code> ; <code>done</code> fires once
<code>tb_param_write.sv</code>	4	Gamma shadow; mid-compute write isolation
<code>tb_power.sv</code>	16	ERR_LOW_BATTERY (0xA); ERR_POWER_FAIL (0xB)
<code>tb_warmup.sv</code>	14	ERR_WARMING_UP (0x8); ERR_INTERRUPTED (0x9)

7.2 Level 2 — Integration Tests (cocotb)

Table 9: Level 2 cocotb integration tests.

Test	Sim time	Coverage
<code>test_reset_outputs</code>	120 ns	All outputs deasserted after reset
<code>test_param_programming</code>	460 ns	<code>gamma_reg</code> / <code>c_reg</code> Q6.10 write/readback
<code>test_sv_counts_set</code>	140 ns	<code>num_sv[0--4] = [60,45,55,50,40]</code> , sum = 250
<code>test_sv_counts_unequal_stress</code>	160 ns	[100,10,80,40,20] extreme distribution
<code>test_qspi_fifo_load</code>	10,420 ns	256-word feature vector loads without overflow
<code>test_qspi_backpressure</code>	168,000 ns	<code>qspi_ready</code> deasserts; writes rejected
<code>test_default_gamma_fixed_point</code>	140 ns	Default $\gamma = 0.25$ (= 0x0100 in Q6.10)
<code>test_full_pipeline_small_batch</code>	71,640 ns	2 SV $\times$ 5-class; <code>done</code> fires, no error
<code>test_kernel_output_range</code>	71,640 ns	All $K \in [0, 1]$ (RBF property holds)

7.3 Level 5 — SPI Co-simulation (m6)

The m6 interface replaces the Wishbone bus with SPI. `tb_spi_cosim.py` acts as the nRF52840 host MCU:

1. Writes 500 alpha coefficients via `ALPHA_WR` (500 SPI frames).
2. Writes gamma, C , and five biases via `PARAM_WR` (7 frames).
3. Preloads SV matrix and input matrix into a Python SRAM model.
4. Fires `CONTROL[start=1, vbatt_ok=1]`.
5. Captures `class_out[2:0]` on each `sample_rdy` rising edge.

Result: **295/300 correct (98.33%)**, zero quantization flips vs. float.

8 Physical Design

8.1 Technology and Die

Table 10: Physical design specifications.

Parameter	Value
PDK	IHP SG13G2 130 nm BiCMOS
Standard cell library	<code>sg13g2_stdcell</code>
Die area	$2400 \times 2400 \mu\text{m}$
Core area (core macro)	$1400 \times 1400 \mu\text{m}$
Top-level die	$2400 \times 2400 \mu\text{m}$
Routing layers	Metal1–Metal5 (<code>RT_MAX_LAYER=Meta15</code>)
Support vectors	500 ([95, 95, 95, 120, 95], VT-boosted)
Total standard cells (top)	157,991
Supply voltage	1.2 V (core), 1.8 V / 3.3 V (IO ring)
Clock	40 MHz (25 ns period)
Clock gate cell	<code>sg13g2_d1clkp_1</code> (ICG)
SRAM	Off-chip IS62WV51216 (1 MB, 10 ns access, async)
Place-and-route tool	LibreLane 3.0.4 / OpenROAD inside Apptainer SIF
HPC cluster	Orca (Portland State University), SLURM

8.2 Pad Assignment (54 pads)

Table 11: Selected pad assignments (power and key signal pads).

Pad	Signal	Direction	Notes
P1	VDD	Power	1.2 V core supply
P2	VSS	Power	Ground
P3	VDDIO	Power	1.8/3.3 V pad ring
P4	VSSIO	Power	Pad ring ground
P5	<code>clk</code>	Input	40 MHz system clock
P6	<code>rst_n</code>	Input	Active-low async reset
P7	<code>spi_csn</code>	Input	SPI chip select (active low)
P8	<code>spi_sclk</code>	Input	SPI clock (CPOL=0)
P9	<code>spi_mosi</code>	Input	MSB-first data in
P10	<code>spi_miso</code>	Output	Read-back
P11–P29	<code>ram_addr[18:0]</code>	Output	SRAM row/column address
P30	<code>ram_ren_out</code>	Output	SRAM read enable
P31–P46	<code>ram_rdata_in[15:0]</code>	Input	16-bit SRAM read data
P47–P49	<code>class_out[2:0]</code>	Output	Predicted class (0–4)
P50	<code>sample_rdy</code>	Output	IRQ: beat result ready
P51	<code>done</code>	Output	IRQ: full batch complete
P52	<code>error</code>	Output	Error flag
P53	<code>error_code[0]</code>	Output	Error code LSB
P54	<code>irq_sample_rdy</code>	Output	Alternate IRQ line

9 Timing Analysis

9.1 Process Corners

The IHP SG13G2 PDK defines three nominal signoff corners:

Table 12: IHP SG13G2 process corners used for signoff.

Corner name	Voltage	Temperature	Condition
<code>nom_typ_1p20V_25C</code>	1.20 V	+25°C	Typical (TT)
<code>nom_slow_1p08V_125C</code>	1.08 V	+125°C	Worst-case setup (SS)
<code>nom_fast_1p32V_m40C</code>	1.32 V	−40°C	Worst-case hold (FF)
<code>fast_1p65V_m40C[†]</code>	1.65 V	−40°C	Extreme hold (outside op range)

[†]Nominal operating range: 1.08–1.32 V. The 1.65 V corner is 37.5% above nominal and not a shuttle requirement.

9.2 Post-PnR Timing Results

Results from OpenROAD `stapostpnr` (step 78, `svm_top_ihp`, jobs 94594/94595):

Table 13: Post-PnR worst-case slack (ns) — all three signoff corners. $\text{WNS} \geq 0$ implies timing met; $\text{TNS} = 0$ implies no violations.

Corner	Setup (max)		Hold (min)	
	WNS (ns)	TNS (ns)	WNS (ns)	TNS (ns)
TT 1.20 V +25°C	0.0	0.0	0.0	0.0
SS 1.08 V +125°C	0.0	0.0	0.0	0.0
FF 1.32 V −40°C	0.0	0.0	0.0	0.0
FF 1.65 V −40°C [†]	—	—	−0.08	0.0

[†]One hold-violating path at extreme over-voltage; 1 path, $\text{TNS}=0$ (path count). Not a shuttle requirement. Setup slack at TT = +12.93 ns (timing closed with > 12 ns margin).

All three required signoff corners pass with **WNS = 0 and TNS = 0**. The design operates comfortably within the 25 ns clock period at all process extremes.

9.3 IR Drop

Post-PnR IR drop analysis (OpenROAD PSM, nominal TT 1.20 V):

$$\Delta V_{\text{GND}} = 0.57\% \times 1.20 \text{ V} = 6.84 \text{ mV} \quad (< 5\% \text{ threshold: MET}). \quad (13)$$

Power-source locations were provided to PSM via `vsrvc_VDD.txt` and `vsrvc_VSS.txt` derived from the pad ring.

10 Power Analysis

10.1 Gate-Level Power Estimate

OpenROAD reports clock-gated, post-PnR power at default activity (12.5% toggle rate at 40 MHz):

Table 14: Post-PnR power (mW) at three corners — `svm_top_ihp` (OpenROAD `stapostpnr`, step 78).

Corner	Internal	Switching	Leakage	Total
TT 1.20 V +25°C	0.657	0.165	0.224	1.046
SS 1.08 V +125°C	0.523	0.129	0.057	0.709
FF 1.32 V −40°C	0.809	0.207	0.830	1.846

10.2 Active Inference Power Estimate

The m6 average power is estimated by scaling the *measured* m5 sky130A duty-cycled average (LAT=3, 500 SVs, 80 bpm: 0.852 mW) by the supply voltage ratio:

$$P_{\text{avg,core}} \approx P_{\text{m5,avg}} \cdot \left(\frac{V_{\text{IHP}}}{V_{\text{sky130}}} \right)^2 = 0.852 \text{ mW} \times \left(\frac{1.2}{1.8} \right)^2 = 0.852 \times 0.444 \approx 0.378 \text{ mW}. \quad (14)$$

This estimate preserves the measured m5 duty cycle (1.29% at LAT=3, 80 bpm) and accounts only for the V^2 supply reduction. Estimated active power: 0.378 mW/0.0129 $\approx$ 29 mW.

10.3 System Power Budget

Table 15: System-level power budget at 80 bpm wearable operation.

Subsystem	Active power	Duty cycle	Avg power
SVM core (500 SVs, LAT=3)	$\approx 29 \text{ mW}$	1.29%	$\approx 0.38 \text{ mW}$
SPI slave + register file	$\approx 0.5 \text{ mW}$	0.1%	$\approx 0.001 \text{ mW}$
ECG analog frontend	$\approx 0.5 \text{ mW}$	100%	0.5 mW
nRF52840 (BLE + MCU)	$\approx 5 \text{ mW}$	2%	$\approx 0.1 \text{ mW}$
Total	—	—	$\approx$ 0.98 mW

Battery budget (200 mAh @ 3.7 V = 740 mWh):

$$t_{\text{battery}} = \frac{740 \text{ mWh}}{0.98 \text{ mW}} \approx 755 \text{ h} \approx 31.5 \text{ days}. \quad (15)$$

The 14-day target is met with $2.25\times$ margin. Compared to the m5 sky130A design (0.869 mW average at the core alone), the IHP SG13G2 version achieves an estimated **40% active power reduction** at the core due to the lower supply voltage.

11 Physical Signoff

Table 16: Physical signoff summary — `svm_top_ihp`, jobs 94594/94595.

Check	Result	Details
Magic DRC	0 violations	<code>drc.magic.rpt</code> : COUNT 0
KLayout DRC	0 violations	<code>drc.klayout.lyrdb</code> : no items
KLayout XOR	0 differences	All layers; Magic vs. KLayout GDS agree
Routing DRC	0 violations	After DRT convergence
Antenna violations	0	Diodes inserted by LibreLane
IR drop (VGND)	0.57%	< 5% threshold (MET)
Setup timing (TT)	MET	WNS = 0, TNS = 0; slack = +12.93 ns
Hold timing (TT)	MET	WNS = 0, TNS = 0
Setup timing (SS)	MET	WNS = 0, TNS = 0
Hold timing (FF)	MET	WNS = 0, TNS = 0
LVS	Functional match	Filler/decap cell mismatch only (PnR artifact)
GDS	Generated	<code>svm_top_ihp.gds</code> (171 MB)
LEF	Generated	<code>svm_top_ihp.lef</code>
Gate-level netlist	Generated	<code>svm_top_ihp.v</code>

11.1 LVS Note

Netgen LVS reports a mismatch for filler and decap cells (`sg13g2_fill_1/2`, `sg13g2_decap_4/8`): these cells have VDD/VSS pins in the LibreLane-extracted layout SPICE but appear as pinless black boxes in the gate-level Verilog, because the PnR tool inserts them post-synthesis and they do not pass through Yosys. Cell pin lists are marked equivalent by Netgen; device counts match ($1258 = 1258$ for top). This is a standard PnR artifact, not a functional mismatch, and is accepted by IHP shuttle teams.

11.2 KLayout XOR

KLayout XOR compares the GDS produced by Magic’s stream-out against the GDS produced by KLayout’s stream-out on a layer-by-layer basis. A non-zero result would indicate a discrepancy between the two GDS writers, suggesting a corrupted or non-physical layout. Both `svm_compute_core` and `svm_top_ihp` produced:

Total XOR differences = 0 (all layers).

12 Comparison with m5 (sky130A)

Table 17: m5 (sky130A / Caravel) vs. m6 (IHP SG13G2 standalone).

Parameter	m5 (sky130A)	m6 (IHP SG13G2)
Process node	180 nm	130 nm
Supply voltage	1.8 V	1.2 V
Interface	Wishbone @ 0x3000_0000	SPI CPOL=0 CPHA=0
Total SVs	500 [95, 95, 95, 120, 95]	500 [95, 95, 95, 120, 95]
alpha_addr width	9-bit	10-bit
Die	Caravel fixed $2920 \times 3520 \mu\text{m}$	Standalone $2400 \times 2400 \mu\text{m}$
Accuracy (Q6.10)	97.67% (293/300)	98.33% (295/300)
Active power (est.)	66 mW	≈ 29 mW
Avg power @ 80 bpm	0.852 mW	≈ 0.38 mW
Inference time (LAT=3)	≈ 9.9 ms/beat	≈ 9.9 ms/beat
Setup WNS	+7.83 ns	+12.93 ns
Cosim testbench	<code>tb_wb_cosim.py</code>	<code>tb_spi_cosim.py</code>

13 System Integration

13.1 Host MCU: nRF52840

Table 18: nRF52840 integration summary.

Feature	Spec	Rationale
Core	ARM Cortex-M4F	Low power; sufficient for feature extraction
BLE	5.0 integrated	Wireless result logging
SPI master	Up to 32 MHz	Drives ASIC SPI at ≤ 2 MHz with margin
Power	4.6 mA @ 3.3 V	Compatible with 200 mAh LiPo
GPIO	48	Enough for SPI + SRAM bus + ASIC IRQ
SDK	Zephyr RTOS / nRF5 SDK	SPI transactions mirror <code>tb_spi_cosim.py</code>

13.2 Model Reload Protocol

The ASIC is fully runtime-reprogrammable. A new trained SVM can be loaded over SPI without resetting the device. The off-chip SRAM (256 KB for the SV matrix) and the on-chip alpha table (1.0 KB) are overwritten in place while the FSM is idle (`STATUS[done]=1`):

1. Write SV matrix to SRAM via MCU direct SRAM bus ($500 \times 256 \times 16$ -bit).
2. Write 500 alpha coefficients via `ALPHA_WR` (`[25 : 16]=SV index`, `[15 : 0]=Q6.10  $\hat{\alpha}$`).
3. Write γ , C , and biases $b_0, \dots, b_4$ via `PARAM_WR`.
4. Optionally update `NUM_SV[0--4]` and `NUM_SAMPLES` (sticky).
5. Assert `CONTROL[start=1, vbatt_ok=1]`.

Constraints: $|S| \leq 500$; $\gamma \in [0, 63.999]$ (Q6.10); $\hat{\alpha} \in [-32, +31.999]$ (Q6.10). Do not write `ALPHA_WR` during active inference.

14 Submission Status

Table 19: IHP shuttle submission checklist.

Item	Status	Notes
Core harden (<code>svm_compute_core</code>)	DONE	Job 94594, 2:03 runtime
Top harden (<code>svm_top_ihp</code>)	DONE	Job 94595, 4:31 runtime
Magic DRC = 0	DONE	Both core and top
KLayout DRC = 0	DONE	Both core and top
KLayout XOR = 0	DONE	Jobs 94594/94595
Setup timing (all 3 corners)	DONE	WNS = 0, TNS = 0
Hold timing (all 3 corners)	DONE	WNS = 0, TNS = 0
IR drop < 5%	DONE	0.57%
GDS on GitHub	DONE	<code>ihp-sg13g2-svm</code> repo
IHP submission issue	OPEN	IHP-GmbH/Open-Silicon-MPW#40
LVS (functional)	DONE	Filler mismatch is PnR artifact
Fast-corner hold (1.65 V)	INFO	WNS = -0.08 ns; outside op range

Shuttle deadline: **August 23, 2026**. GitHub submission repository: <https://github.com/adamleehandwerger/ihp-sg13g2-svm>.

15 Error Code Reference

Table 20: Error codes reported in `STATUS[5:2]`.

Code	Name	Condition
0x1	<code>ERR_SV_ZERO</code>	All <code>num_sv[k]</code> = 0 at start
0x2	<code>ERR_SV_OVERFLOW</code>	Total SV count exceeds <code>alpha_table</code> depth
0x3	<code>ERR_FIFO_OVERFLOW</code>	Feature FIFO overflow (too many SRAM reads)
0x6	<code>ERR_GAMMA_SAT</code>	$\gamma \cdot D$ overflows LUT address width
0x7	<code>ERR_GAMMA_ZERO</code>	$\gamma_{\text{int}} = 0$ at start (all kernels = 1.0)
0x8	<code>ERR_WARMING_UP</code>	Advisory: ≤ 100 beats since cold start; auto-clears
0x9	<code>ERR_INTERRUPTED</code>	Advisory: <code>rst_n</code> during warm-up window; auto-clears
0xA	<code>ERR_LOW_BATTERY</code>	Advisory: <code>vbatt_warn</code> asserted (device continues)
0xB	<code>ERR_POWER_FAIL</code>	Blocking: <code>vbatt_ok</code> deasserted (start blocked)

Codes 0x1–0x7 are *sticky* (latched until `rst_n`). Codes 0x8–0xB are *non-sticky* advisories that track the live pin or state. A real fault (code < 0x8) overrides any advisory in the `STATUS` register.

A Key File Locations

Location	Path
ECE410 repo	https://github.com/adamleehandwerger/ECE410
Submission repo	https://github.com/adamleehandwerger/ihp-sg13g2-svm
RTL (m6)	project/m6/rtl/{compute_core.sv, top.sv, interface.sv}
Core config	project/m6/synth/core_config.json
Top config	project/m6/synth/top_config.json
Harden scripts	project/m6/pnr/{core_harden.sh, top_harden.sh}
SPI cosim	project/m6/tb/tb_spi_cosim.py
Orca artifacts	/scratch/funphin-openlane_svm/svm_m6_artifacts/
GDS (top)	svm_top_ihp.gds (171 MB)

Acknowledgments

Computing resources. All place-and-route, timing signoff, and cosimulation jobs were run on the *Orca* high-performance computing cluster at Portland State University (PSU Research Computing), using LibreLane 3.0.4 inside an Apptainer container on the long SLURM partition.

Data. The cardiac arrhythmia classifier was trained and evaluated using the following publicly available resources:

- **MIT-BIH Arrhythmia Database** (PhysioNet record nos. 100–234): G. B. Moody and R. G. Mark, “The impact of the MIT-BIH Arrhythmia Database,” *IEEE Eng. Med. Biol. Mag.*, vol. 20, no. 3, pp. 45–50, 2001.
- **PhysioNet / PhysioBank / PhysioToolkit**: A. L. Goldberger *et al.*, “PhysioBank, PhysioToolkit, and PhysioNet,” *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000. <https://physionet.org>
- **IHP SG13G2 Open-Source PDK**: IHP Microelectronics (Leibniz Institut für innovative Mikroelektronik), *ihp-sg13g2 Open-Source PDK*, Frankfurt (Oder), Germany, 2024.
- **LibreLane EDA flow**: Efabless / The OpenROAD Project, *LibreLane 3.0.4*, <https://github.com/efabless/openlane2>, 2024.

B References

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- [3] M. Llamedo and J. P. Martinez, “Heartbeat classification using feature selection driven by database generalization criteria,” *IEEE Transactions on Biomedical Engineering*, vol. 58, no. 3, pp. 616–625, 2011.

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